

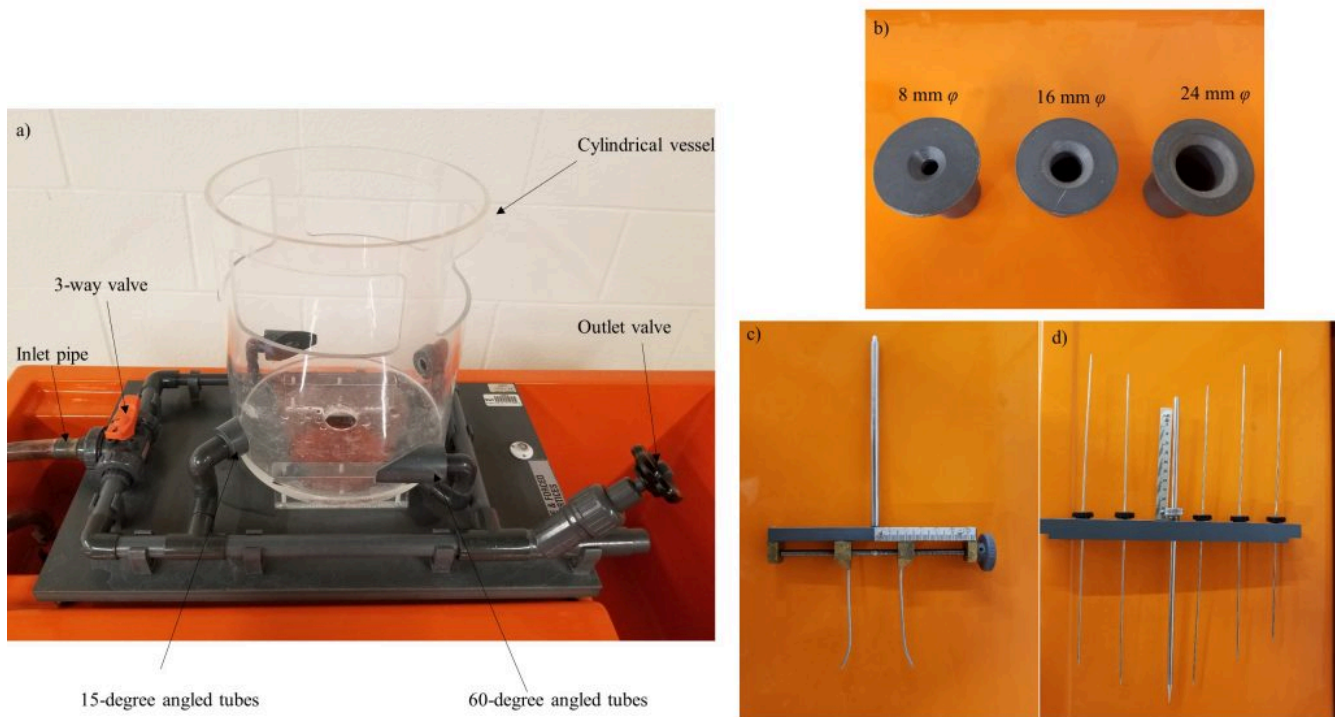


Figure 8.1: a) P6238 CUSSONS free and forced vortex apparatus, b) push-in orifices, c) free vortex measuring caliper, d) force vortex measuring probes

7. THEORY

Two types of vortices are distinguished in the dynamics of the motion: forced and free vortices. The forced vortex is caused by external forces on the fluid, such as the impeller of a pump, and the free vortex naturally occurs in the flow and can be observed in a drain or in the atmosphere of a tornado.

7.1. FREE VORTEX

A free vortex is formed when water flows out of a vessel through a central hole in the base (Figure 8.2). The degree of the rotation depends on the initial disturbance. In a free cylindrical vortex, the velocity varies inversely with the distance from the axis of rotation (Figure 8.3).

$$v = \frac{k}{r} \quad (1)$$

The equation governing the surface profile is derived from the Bernoulli's theorem:

$$\frac{v^2}{2g} + z = C \quad (2)$$

Substituting Equation (1) into (2) will give a new expression:

$$\frac{k^2}{2gr^2} + z = C \quad (3)$$

or: